

Random Effects and Linear Mixed Models

Part IV treated the levels of a factor as chosen: five nitrogen rates, three machines, the treatments the experimenter cared about. Many factors are not like that. The twelve laboratories in a proficiency study, the eleven firms in a panel, the patients in a trial and the classrooms in a school district are a *sample* of levels, and the question is not how laboratory 3 compares with laboratory 7 but how much laboratories vary. Declaring such a factor random changes three things at once. The parameters of interest become variances. The individual levels are *predicted* rather than estimated, and the prediction is shrunk towards the overall mean. And the covariance of the response stops being $\sigma^2\mathbf{I}$: observations sharing a level are correlated, which is what Part VII is about. This chapter develops the model that results, the best linear unbiased predictor and Henderson's equations that compute it, restricted maximum likelihood, and what inference survives when the covariance itself has been estimated from the same data.

What you need. [Chapter 6](#) (projections and generalized least squares, Definition 6.9.1, Theorem 6.9.3), [Chapter 7](#) (the Gauss–Markov and Aitken theorems, Theorem 7.2.1 and Corollary 7.2.3; maximum likelihood, Theorem 7.3.1; the conjugate posterior, Theorem 7.6.2), [Chapter 4](#) (distributions of quadratic forms, Theorem 4.3.2 and Theorem 4.4.2), [Chapter 9](#) (expected mean squares, Theorem 9.6.1 and Exercise (B1)), [Chapter 14](#) (best linear unbiased prediction, Theorem 14.2.8, resting on the best linear predictor Theorem 2.5.2 of [Chapter 2](#)), [Section 15.1](#) (the one-way layout and the first look at random effects, Definition 15.1.1) and [Chapter 16](#) (nested and crossed layouts, Theorem 16.5.1). From Part V, [Section 21.3](#) (weighted least squares with estimated weights) and [Section 23.3](#) (the parametric bootstrap, Proposition 23.3.3); from Part VI, [Chapter 27](#) (ridge and empirical Bayes, Theorem 27.2.2 and Proposition 27.5.4) and [Chapter 29](#) (AIC and BIC, Definition 29.2.2). The generalized least squares theory of [Chapter 31](#) is used throughout. Underneath the whole chapter sits the partitioned-matrix algebra of [Chapter 1](#) — the Woodbury identity, the block determinant, the partitioned inverse, the cyclic trace, a matrix square root and the calculus of quadratic forms with Lagrange multipliers (Theorem 1.6.5, Eq. 1.6.6, Theorem 1.6.3, Theorem 1.6.4, Theorem 1.6.1, Theorem 1.7.6, Proposition 1.11.2, Proposition 1.11.3) — which is what turns Henderson's equations and the likelihood into something computable; and, in [Section 32.4](#), the normal density and its behaviour under a linear map from [Chapter 3](#) (Theorem 3.2.2, Theorem 3.3.1). [Section 32.3](#) points forward to the smoothing spline as a mixed model ([Section 43.6](#)); that is a promise, not a prerequisite.

32.-1.1 Roadmap

- [Random effects](#)
- [The linear mixed model](#)
- [Prediction: BLUP and Henderson's equations](#)
- [Maximum likelihood and REML](#)
- [Inference when the covariance is estimated](#)
- [Bayesian thread III: mixed models as hierarchical models](#)
- [Summary and notes](#)